# The community ecology of cancer: From eco-evolutionary dynamics to therapeutic strategies

## Specific objectives

Traditionally, cancer research has focused on cell-intrinsic genomic changes as the mechanistic basis of tumor initiation and progression. A mutant cell that breaks free of the regulatory mechanisms limiting its growth suddenly finds itself in a hostile but resource-accessible environment. If it can establish and start to proliferate, the basic principles of Darwinian evolution will apply. And, because cancer cells are notorious for their high mutation rates, one can expect malignant lineages to rapidly adapt to their environment and diversify as they do so.

**This project starts from the complementary perspective that a tumor is a miniature ecosystem.** The composition of cancer clones is organized by the ecological opportunities available within the host [1]. **Coexistence mechanisms that ecologists routinely study in forests, lakes, and microbial systems also apply to tumors**: one finds resource partitioning (different clones consuming different nutrients), proliferation-survival tradeoffs, fluctuation-dependent coexistence (clones that tolerate temporary stress outlast those that cannot), and spatial niche differentiation across the tumor microenvironment [2].

Figure 1 illustrates this key intuition in a laboratory image from colonic adenocarcinoma [3]. In the same tumor, cancer cells at the tumor-stroma interface are exposed to immune cells and thus upregulate immune suppression programs, whereas cells in the tumor nest do not. Yet neither clone dominates the other—instead, the two stably coexist despite their direct proximity. What we have here is textbook ecological niche differentiation, where distinct clones specialize in either the vascularized, resource-rich but immune-challenged periphery, or the hypoxic but protected core. This perspective turns intratumoral heterogeneity from an uninterpretable mass of random genetic variants into a structured ecosystem. **By uncovering the rules which govern the tumor ecosystem, we can devise ecologically-based interventions for dismantling it.**

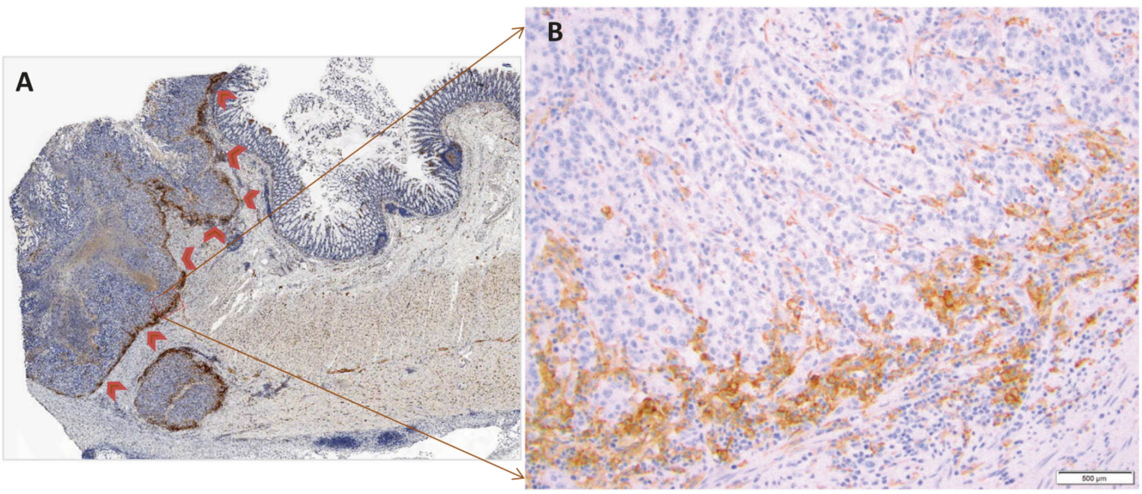


**Figure 1:** Immunohistochemical images of the invasive front of colonic adenocarcinoma (**A**, with the small boxed region magnified in **B**). Cells were immunostained for PD-L1, an immune checkpoint ligand. Only those cells near the tumor-stroma interface (outlined by the red arrows), and therefore regularly exposed to immune cells, express PD-L1. Figure taken from Liu et al. [3].

In light of this approach, the proposal asks **which ecological mechanisms stabilize coexisting cancer phenotypes, and how we can disrupt those mechanisms to target the tumor ecosystem as a whole**. The central hypothesis is that lasting therapeutic results require **community-level** therapy. Rather than repeatedly eliminating whichever clone is currently dominant in a whack-a-mole style game, the project will identify the coexistence mechanisms that maintain diversity, and devise strategies for breaking them. The expected result is controlled tumor homogenization—after which conventional therapies can act more efficiently.

To do this, the four work packages (WPs) of the project are organized as an iterative workflow from theory to empirical validation and back. WP1 builds mechanistic coexistence models using ecological theory and adaptive dynamics. WP2 stress-tests them in a more realistic but still theoretical digital evolution setting. WP3 translates the mechanisms into intervention strategies. WP4 tests predictions against tumor data and laboratory experiments, feeding the results back into WP1.

**The overarching purpose of this project is to establish a predictive community ecological framework for understanding intratumoral heterogeneity, and to use this knowledge to design robust, testable, ecologically informed therapeutic strategies.** The specific aims are:

1. **Understand the mechanisms maintaining intratumoral diversity (WP1):** Build mechanistic mathematical models of tumor clone eco-evolutionary dynamics, from ecologically motivated tradeoffs (e.g., growth-migration, immune evasion-metabolic efficiency, aerobic-anaerobic metabolism).
2. **Make general predictions on the coexistence of tumor clones (WP1-WP2):** Derive and stress-test predictions about which environmental and physiological conditions promote, maintain, or suppress intratumoral diversity.
3. **Design therapeutic strategies (WP3):** Translate the theoretical results into two novel therapeutic classes grounded in ecological mechanisms: **community disassembly** (targeted homogenization followed by eradication) and **ecological cascade therapy** (leveraging indirect interactions to collapse the tumor community).
4. **Empirically validate and refine results (WP4):** Test predictions using tumor cross-section data and laboratory experiments, then iteratively refine models and interventions based on their outcome.

## Background

It is well established that solid tumors (comprising around 90% of all tumors [4]) exhibit extensive genetic and phenotypic diversity [5, 6], and cancer biology has made great progress in characterizing the molecular features that underlie this heterogeneity. The classic "hallmarks of cancer" framework elegantly describes the capabilities that tumor cells acquire [7, 8], while evolutionary theory has illuminated how phenotypically distinct clones determine tumor composition [9–11].

By contrast, an eco-evolutionary perspective on interpreting tumor clone coexistence has been a much more recent development [1]. An important insight this line of research has provided is that **the ecological and evolutionary tradeoffs tumor cells face have strong analogies with those studied in population and community ecology** [2]. Examples include competition-colonization [12] and competition-mortality [13] tradeoffs where clones are either locally competitive or proliferate faster [14–16], tradeoffs induced by temporal fluctuations in nutrient supply [17], and metabolic tradeoffs favoring either oxidative phosphorylation (OXPHOS) or aerobic glycolysis (the Warburg effect)—analogous to ecological strategies of efficient resource use vs. rapid but wasteful consumption [18].

If tumors can indeed be understood through the lens of ecological tradeoffs, it is a natural next step to map out the space of ecological opportunities induced by them, and to use the toolbox of coexistence theory and community stability analysis to understand how one could undermine cancer cell diversity. However, these steps are still lacking—developing them is precisely the purpose of this project. Fortunately, community ecology has a rich set of tools applicable in this context, many of which I have directly contributed to developing. Examples include methods for determining the sensitivity of communities to environmental or evolutionary perturbations [19–22], understanding factors (dis-)allowing the stable coexistence of competing populations [23, 24], structural stability analysis (robustness of a community to parameter changes) [25, 26], predicting how established communities respond to the arrival of new competitors [27], and understanding the stability of diverse multi-population assemblages [28, 29].

In addition to these analytical tools, digital evolution systems such as Avida [30] use self-replicating computer programs to study evolutionary dynamics *in silico*. This method allows complex ecological and evolutionary dynamics to emerge in more complex, realistic environments, complementing the previous analytical-mathematical methods. Digital evolution platforms offer perfect experimental control and repeatability: every aspect of the environment can be specified and monitored, and because any state of the system can be saved, rewound, and replayed, even destructive perturbations are fully reversible. These properties make digital evolution a powerful bridge between mathematical models and empirical systems [31]. Indeed, digital evolution has been used to fruitfully study problems such as the origin of complex adaptations [32], the emergence of altruism [33], and has even found ecological applications [31, 34, 35].

This project will move beyond what is currently known by applying these tools in the context of tumor diversity, to develop improve, ecologically informed therapies. **By identifying the mechanisms that stabilize coexistence across tumor clones, we can pinpoint the precise environmental interventions required to dismantle them.**

## Work plan and preliminary results

### Theory and method

The project is organized into four work packages that follow a logical progression from theoretical foundations to empirical validation. WP1 develops analytical models of tumor clone coexistence. WP2 validates and extends these predictions using digital evolution. WP3 translates the resulting ecological understanding into therapeutic strategies. WP4 tests predictions against real tumor data, which then feeds back to WP1 to inform refined models—and so on, in a cycle of iterative improvements.

**WP1: Analytical models of intratumoral coexistence.** The goal of this WP is to develop useful mathematical models of tumor dynamics characterized by various intratumoral tradeoffs. These models will quantify general mechanisms for the coexistence of phenotypically distinct cancer clones. "Useful", in this context, means that the models make testable predictions about what tumoral variation one should expect to arise, and how this diversity depends on the host's microenvironmental and physiological conditions. We will systematically explore a suite of tradeoffs typically hypothesized to be important in tumor evolution [2]:

- **Go-or-grow** (competition-colonization tradeoff): Clones either reproduce efficiently and are locally competitively dominant, or are superior dispersers—potentially seeding metastases [14].
- **Immune evasion vs. metabolic efficiency**: Upregulating immune checkpoint ligands or other immune evasion mechanisms allows access to vascularized, nutrient-rich zones, but at a metabolic cost.
- **Metabolic plasticity** (growth rate vs. substrate flexibility): Some clones proliferate rapidly on glucose, others survive on alternative substrates such as glutamine or lactate.
- **Habitat selection** (vascular vs. hypoxic zones): Clones specialize on normoxic, nutrient-rich perivascular regions or survive in hypoxic, acidic tumor interiors.

- **Temporal niche partitioning**: Rapid aerobic glycolysis (Warburg-type metabolism) vs. efficient oxidative phosphorylation during periods of nutrient scarcity.

For each tradeoff, we will first employ adaptive dynamics [36, 37], a mathematical framework for modeling how new phenotypes arise through sequential mutations and either establish or go extinct, to simulate the evolutionary diversification of tumor cells from a single ancestor. This approach, which assumes clonal reproduction consistent with tumor biology, iteratively introduces rare mutants with slightly altered trait values to generate emergent, multi-phenotype communities. Once these candidate communities are established, we will apply coexistence theory [19, 23, 24] and community stability analysis [20–22, 25–29] (quantitative tools for determining what maintains or destabilizes multi-clone communities) to understand the factors responsible for stabilizing them, and, crucially, how one can best disrupt that stability.

Importantly, studying the above tradeoffs in isolation is just the first step. Once we have a good understanding of them individually, **we will move on to models that combine multiple tradeoffs simultaneously**, using similar methods to determine how various tradeoff combinations support intratumoral diversity, and to identify the weak points of these mechanisms through which they can best be dismantled.

**WP2: Validation using digital evolution.** Analytical models necessarily simplify biological complexity. WP2 will test whether the coexistence mechanisms and predictions derived in WP1 hold when ecological and evolutionary dynamics are allowed to emerge without specifying what traits can evolve and how, thus conforming to Orgel's Second Rule.[1] It also checks the robustness of the proposed mechanisms against random fluctuations in cell birth and death events as well as microenvironmental variability (the former is a natural byproduct of digital evolution systems that work via individual-based simulations; the latter is optional but simple to add). We will implement digital evolution experiments using the Avida platform [30], where populations of self-replicating digital organisms evolve under selective pressures mimicking the tumor microenvironment. This WP will greatly benefit from participating researcher Emily Dolson (Michigan State University) who is an expert in digital evolution systems and their ecological applications in particular [31, 35].

This WP will take the hypotheses and predictions arising from WP1 and check whether they hold up under a more realistic digital evolution setting. The lessons gained from the digital evolution experiments will be used in guiding and refining the models of WP1. **Digital evolution thus serves as a still theoretical, but nevertheless intermediate validation tier between analytical theory and empirical data**: it provides a level of experimental control that is unavailable in real biological systems, and can also capture emergent dynamics that are absent from simplified mathematical models.

**WP3: Developing ecologically informed treatment strategies.** This WP translates the ecological understanding from WP1 and WP2 into suggested therapeutic strategies that can be tested experimentally. The challenge is that ecological model predictions can rarely be implemented directly, owing both to practical laboratory constraints and the imperative to minimize intervention to protect the patient's health. We will explore two main avenues for realizing ecologically informed, minimally invasive therapies.

The first is **community disassembly**. For each coexistence mechanism identified in WP1, we will determine theoretically which environmental parameters, if changed, would break coexistence and lead to competitive exclusion. The parameters we look at will all correspond to targetable interventions such as anti-angiogenic therapy (reducing vascularization heterogeneity), immunotherapy (altering immune pressure), and metabolic inhibitors (shifting the availability

[1] Commonly paraphrased as "Evolution is cleverer than you are"—i.e., evolution can produce creative solutions which are surprising to humans. Clearly, this potential for surprise is greatly diminished in analytical models, where the space of possible adaptations is known at the outset. Digital evolution does not have this drawback [38].

of various resources). We will focus on designing "first strike / second strike" protocols. The first strike is a subtle environmental intervention, just enough to shift the competitive balance so that one or a few clones gain a decisive advantage and drive the rest to extinction, homogenizing the tumor. The second strike then applies conventional therapy to a now uniform, and far more vulnerable, target. Stripped of the phenotypic diversity that enables adaptive resistance, the remaining clone has little room to evolve its way out.

The second avenue is **ecological cascade therapy**. Using the community structures from WP1, we will identify indirect interaction pathways through which the disruption of one clone leads to the decline of another. We will determine the conditions under which perturbing a clone that is perhaps not so important but is easily treatable will trigger a cascade that eliminates a more resistant one. While this may sound overly optimistic, many ecological systems exhibit precisely the structural properties that make them susceptible to such cascades [20, 22, 39]. Ecological cascade therapy is therefore a structurally grounded strategy, not merely a theoretical hope. To showcase this, a proof-of-concept example is presented under "Preliminary results" below.

**WP4: Empirical validation.** Once WP1 and WP2 yield predictions which are translated into empirically testable ideas by WP3, these will be validated using two complementary empirical approaches. First, we will analyze spatial transcriptomic and histological data available through the labs of Cantù and Stallaert to test whether patterns of clone diversity correlate with microenvironmental variables (hypoxia, vasculature proximity, etc.) as predicted by our models, using spatial point pattern analysis and diversity indices.

Second, we will generate high-resolution spatially resolved maps of cancer heterogeneity using multiplexed immunofluorescence (50–100 biomarkers) and NGS-based transcriptomics. This is made possible by the Stallaert lab's capabilities. We can then use these data to map phenotypes to ecological niches within tumors and develop patient-derived organoid models to test WP3 therapeutic interventions. Additionally, the Cantù lab will analyze freshly extracted human colorectal cancer and hepatocellular carcinoma biopsies at subcellular resolution using 10x Genomics Visium HD combined with NGS transcriptomics, leveraging its expertise in transcription factor biology to identify functional gene regulatory hierarchies given the pattern of gene expression and the extracellular milieu to which each cell is exposed. A key strength is that each biopsied sample is accompanied by histologically normal adjacent tissue, enabling intraindividual comparisons across healthy tissue, primary tumor, and metastatic formations for each patient. This permits testing predictions at both the intraindividual and interindividual levels, distinguishing conserved cellular ecologies from patient-specific patterns driven by distinct mutational landscapes or therapeutic regimens.

**WP4 is a final validation step that grounds the theoretical framework in empirical reality, and identifies which model assumptions need refinement.** Initially, we do not expect the theoretical predictions to be quantitatively correct; rather, having performed the experiments they inspire, we will use the results to update the models and suggest improved therapeutic strategies—which can then be tested in a new round of experiments, closing the iterative loop.

### Time plan and project organization

I (György Barabás) will serve as PI and devote 50% of my time to this project. My expertise in mathematical ecology and coexistence theory is central to WPs 1–3. A postdoctoral researcher in experimental oncology will be hired for 3 years to lead WP4. Ideally, the project will be expanded to include a second postdoctoral researcher in theoretical ecology, contributing to WPs 1–3. This position is contingent upon securing supplementary funding from the Swedish Research Council (VR), for which a parallel application has been submitted.

The project will benefit from the following **core collaborating researchers:**

- **Anuraag Bukkuri** (Lecturer; City St George's, University of London, United Kingdom): Expertise in mathematical oncology and evolutionary game theory applied to cancer. Will contribute to the development and analysis of tumor evolutionary models (WPs 1–3).
- **Claudio Cantù** (Professor; Linköping University, Sweden): Expertise in genome regulation and transcription-factor profiling in development and disease. Prof. Cantù leads an experimental lab of approximately ten researchers and will provide local expertise for the empirical validation in WP4, including access to experimental facilities and guidance on translating theoretical predictions into testable laboratory protocols.
- **Emily Dolson** (Assistant professor; Michigan State University, United States): Expertise in digital evolution and the Avida platform. Will lead the digital evolution experiments (WP2) and contribute to model validation.
- **Wayne Stallaert** (Assistant professor; University of Pittsburgh, United States): Expertise in obtaining detailed, quantitative molecular descriptions of each cell in the tumor microenvironment (both cancer and non-cancer host cells such as fibroblasts, immune cells, endothelial cells, etc.) using highly multiplexed immunofluorescence on sections of human tumors. With these data, the lab can determine how distinct phenotypic states are spatially organized within a given tumor. They also develop methods using patient-derived organoid models to mimic and modulate the native 3D tumor environment *in vitro*. Will contribute to WP4, collecting high-resolution phenotypic diversity data from tumor samples.

We also have a broader network of **additional theoretical collaborators** with whom we are already holding regular meetings on eco-evolutionary cancer modeling, and who have explicitly agreed to participate in this project:

- **Jie Deng** (Postdoctoral researcher; Oxford University, United Kingdom)
- **Gaurav Baruah** (Assistant professor; Indian Institute of Science, Bangalore, India)
- **Yun Tao** (Assistant professor; Stony Brook University, United States)

The project is planned for four years: three funded by Cancerfonden, plus an additional year for theoretical development. The project is structured so that theoretical insights from earlier phases directly inform subsequent validation and strategy development:

- **Year 1: WP1 (Foundations).** Development of single-tradeoff analytical models using adaptive dynamics to generate emergent tumor communities. Initial analysis of coexistence and community structure. Concurrent launch of the Avida digital evolution experiments. (If VR funding is secured: hire theoretical postdoc.)
- **Year 2: WP1 (Synthesis) & WP2 (Validation).** Extension to multi-tradeoff models. Evaluation of digital evolution experiments for validating WP1 predictions. Feeding results back into WP1 for refinement. First publications on theoretical foundations. Experimental postdoc starts, with acquisition of tumor data and testing of model predictions.
- **Year 3: WP3 (Strategies) & WP4 (Data Prep).** Design of community disassembly and ecological cascade therapy protocols. *In silico* testing using analytical and digital frameworks. Experimental postdoc tests model predictions against tumor data.
- **Year 4: WP4 (Empirical Validation) & Synthesis.** Statistical analysis of tumor data to test therapeutic strategy predictions. Iterative refinement of models based on empirical results. Final integration of all work packages and preparation of high-impact publications.

We will conduct **weekly virtual meetings** with the theoretical team (Baruah, Bukkuri, Deng, Dolson, Tao, and me, plus the theoretical postdoc if VR funding is secured) to refine models and interpret simulation results, and **monthly video conferences** with empirical partners (Cantù, Stallaert, and the experimental oncology postdoc) to align experimental design with theoretical predictions. We also plan to host **two in-person workshops in Linköping, Sweden** to foster interdisciplinary exchange. Results will be published in peer-reviewed journals and presented at international conferences throughout the project.

Three important **risk factors** are (i) that certain tradeoff models may not support robust coexistence, (ii) that theoretical predictions may seriously diverge from empirical observations, and (iii) that VR funding may not be secured, in which case the theoretical postdoc position would not materialize. The first risk is addressed by exploring a broad suite of tradeoffs in WP1, with WP2 providing an alternative validation route in case analytical models prove insufficient. The second is mitigated by the iterative feedback loop between modeling and experimentation, which allows for rapid model refinement regardless of the initial outcome. The third is acceptable because the PI and our strong team of collaborating theoreticians can carry out WPs 1–3 even without a dedicated theoretical postdoc.

**Preliminary results**

To illustrate the core logic of the project, here I present a proof-of-concept example based on a competition-mortality tradeoff [2, 12, 13] in which the speed of cancer cell proliferation is an evolving trait. Cells that prioritize rapid division incur metabolic costs in the form of reduced oxidative phosphorylation efficiency [15], rendering them inferior competitors for limited resources [7]. I implemented this model [12, 13] in an eco-evolutionary simulation that numerically integrates the ecological dynamics, periodically interrupting the integration to introduce a new random mutant phenotype into the community. Starting from a single phenotype and allowing evolution to proceed freely, the system had diversified into a stable community of seven coexisting phenotypes by around mutation event 7000, and this community persisted for the remainder of the simulation (Figure 2).

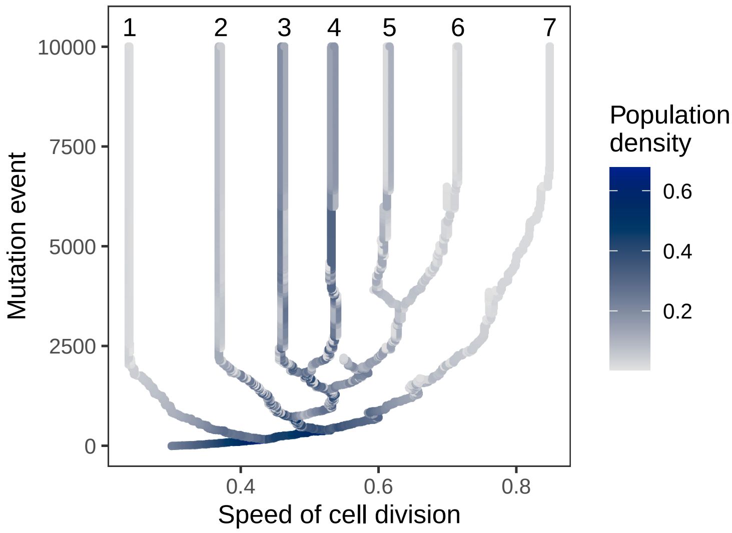


**Figure 2:** Starting from a single phenotype, the system under a competition-mortality tradeoff [12, 13] diversifies into a community of seven coexisting tumor clones. The community reaches an eco-evolutionary stable state [40] around mutation event 7000. Each point represents a clone with a distinct phenotype (x-axis) at a given stage of the evolutionary simulation (y-axis; measured in mutation events), with colors indicating the clone's population density.

Given the ecological interactions among these seven stably coexisting clones, we can determine many useful facts about the structure of this community. For example, one can construct the *community matrix* [41], which is a table of values summarizing how each clone affects every other one in the short term (Figure 3, left). The table has weak entries whenever the column-clone has a larger index than the row-clone, and tends to have strong entries otherwise. This indicates

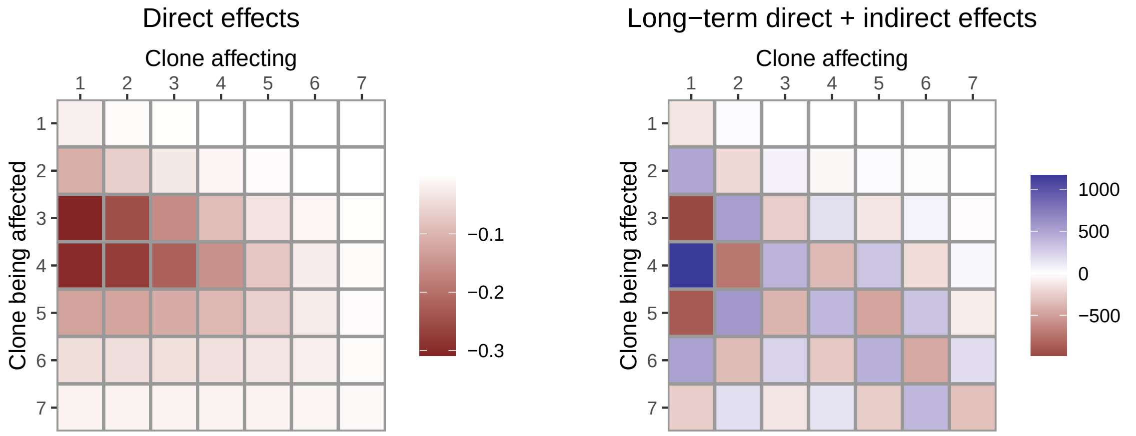


**Figure 3:** Short term direct (left) vs. long-term net effects of clones on one another (right). For example, the entry in row 3 and column 1 measures the effect of a change in clone 1's abundance on clone 3. The clone numbers correspond to those in Figure 2.

hierarchical competition: the top competitor (clone 1) experiences little pressure from others, and each clone is affected primarily by those above it in the hierarchy, not the ones below.

Using these values, one can go further and construct another table which measures not just the immediate direct effects, but the long-term net effect of every clone on every other one [19, 42] (Figure 3, right). Crucially, this table reveals that disturbing the highly competitive but slow-growing first clone has a very strong effect on clones 3, 4, and 5, which are less resource-efficient but proliferate much faster. In particular, an increase in clone 1 will cause a decrease in clones 3 and 5, possibly causing their extinction. **This validates the hypothesis behind ecological cascade therapy**: we need not target the more aggressive, faster-growing clones directly, because we can collapse the community by appropriately manipulating other clones instead. Going further, we can use mathematical methods such as *singular value decomposition* [43] to determine the disturbance to the tumor whose effects get amplified the most. In this example, that disturbance corresponds roughly to positive effects for odd-indexed clones and negative ones for even-indexed clones. Such disturbances will be greatly amplified, suggesting that even modest interventions could induce major shifts in tumor composition.

In the project, predictions such as these will be translated into specific experiments that can be performed in the lab on tumor samples. The experimental results will then inform and feed back into model development, leading to refined models and updated hypotheses to be tested. In this iterative manner, the goal is to eventually settle on truly predictive models which have the ability to guide effective, experimentally validated medical interventions.

## Significance

This project addresses a critical conceptual gap by systematically applying the quantitative tools of community ecology to the problem of intratumoral heterogeneity. Existing literature has noted oncological applications of ecology, but no prior work has rigorously used these tools to predict or manipulate tumor community structure. For instance, one can quantify how sensitive a clone community is against sustained microenvironmental change [19–22], predict whether a new mutant clone will take hold or be excluded [27], and determine whether temporal variability in the microenvironment promotes clonal diversity [23, 24]. Crucially, none of these tools have been used for devising strategies to dismantle tumors.

This approach yields two novel, ecologically grounded therapeutic paradigms. The first is **community disassembly**. It targets the coexistence mechanisms themselves to homogenize the tumor, rendering it vulnerable to a focused second-strike therapy—an approach analogous to evolutionary double-bind strategies [44]. The second is **ecological cascade therapy**, which hijacks indirect ecological dependencies within the tumor. Rather than attacking the most aggressive clone directly, it targets a more accessible one whose removal triggers a cascade of extinctions that collapses the entire community. **These two strategies represent a fundamental departure from current clinical approaches, offering a systemic method to overcome the adaptive resistance that plagues conventional therapies.**

Finally, the project bridges the divide between abstract theory and empirical reality. It establishes a rigorous feedback loop where theoretical models generate testable predictions that are validated against laboratory observations and tumor data. This synthesis lays the groundwork for the new subdiscipline of **tumor community ecology**. If successful, this work will transform how we understand, monitor, and treat cancer, shifting the paradigm from fighting individual mutations to managing the ecological landscape of the tumor itself.

## Independent research line

This project represents a new and independent line of research for me. My previous work has focused on traditional community ecology and eco-evolutionary dynamics. The application of coexistence theory to intratumoral heterogeneity is a novel extension of this expertise. The project is distinct from my previous VR Starting Grant (which focused on species coexistence in changing environments and concluded in 2022) and from my current grant from the Swedish Environmental Protection Agency on wildlife ecology. The cancer ecology direction was initiated through new collaborations formed in 2025-2026 and has not been funded previously.

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